Environmental Sciences & Renewable Energy (ME-306)

Module – 2

- ***Water Resource***
- ***Water Pollution: Definition and Classification***
- ***Sources of Contamination***
- ***Pollutants, Their Types and Effects***
- ***Municipal Water Treatment: Slow and Rapid Sand Filter***
- ***Disinfection: Methods, Advantages & Disadvantages***

Mahesh Chandra Mehta

The Green Avenger of India

Goldman Environmental Prize (1996)

Ramon Magsaysay Award (1997)

Padma Shri (2016)

- ***Shriram Food Fertilizer Case/Oleum Gas Leak Case (1986)***
- ***Taj Trapezium Case (1997)***
- ***Gamma Chamber Case/Safeguard from Radiation Case (1987)***
- ***Ganga Pollution Case/Kanpur Leather Tanneries Case (1988)***
- ***Introduction of CNG /Delhi Vehicular Pollution Case (1991)***
- ***Waste and Hazardous Substances Case (1992)***
- ***Environmental Education Case (1992)***
- ***Ground Water Pollution Case (1989)***
- ***Delhi NCR Pollution Case (2019)***

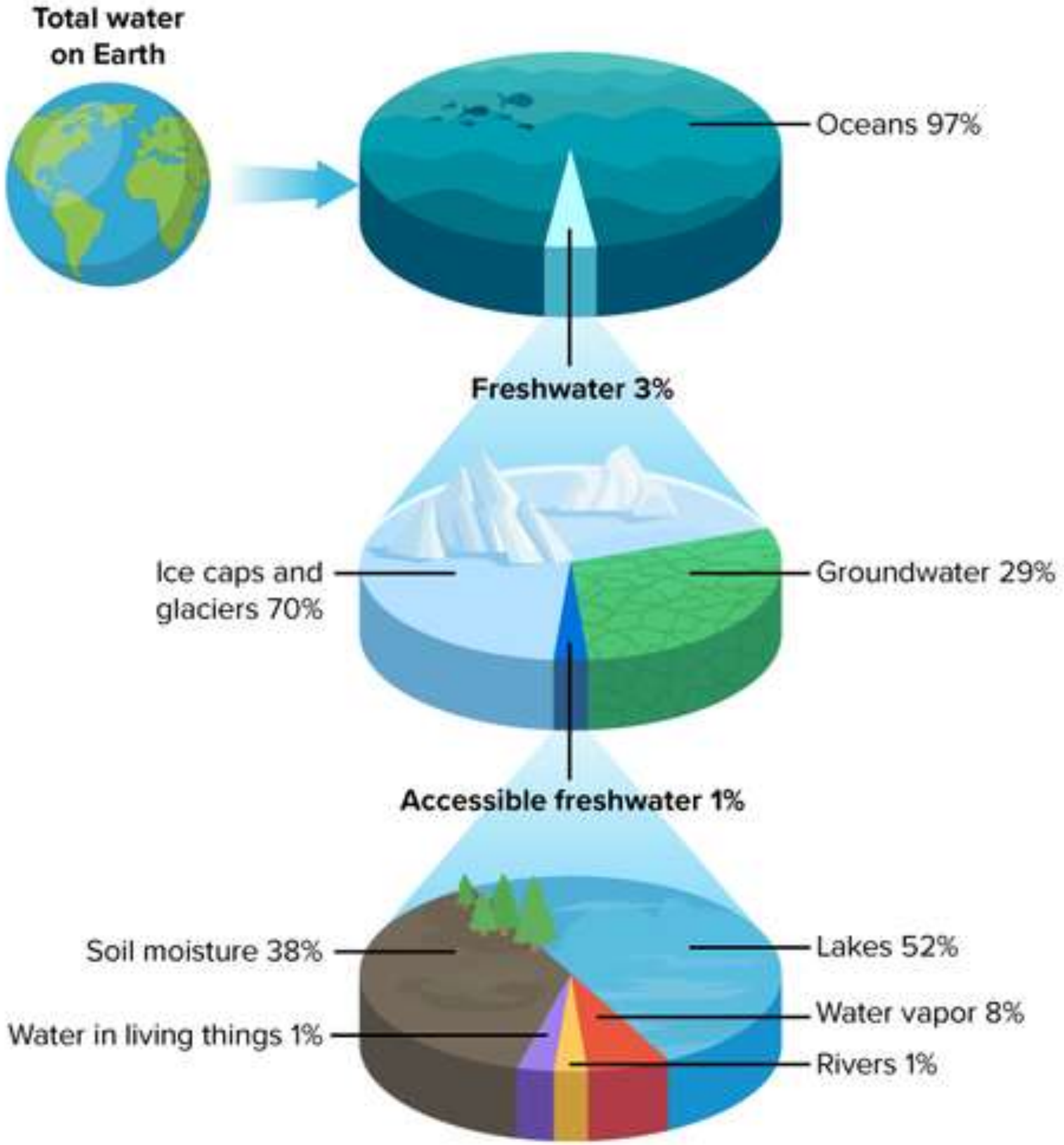
Lecture No. 1

- **Water Resources**
- **Water Pollution and it’s Types**

Water Resources

Water resources are natural sources of water that are potentially useful to humans and other living beings.

- 97% of the water on Earth is salt water, and only three percent is fresh water; slightly over two-thirds of this is frozen in glaciers and polar ice caps.
- The remaining unfrozen freshwater is found mainly as groundwater, with only a small fraction present above ground or in the air.



Water distribution on the Earth

Sources of water *can be broadly classified into two categories*:

- ***Surface water***
- ***Groundwater***

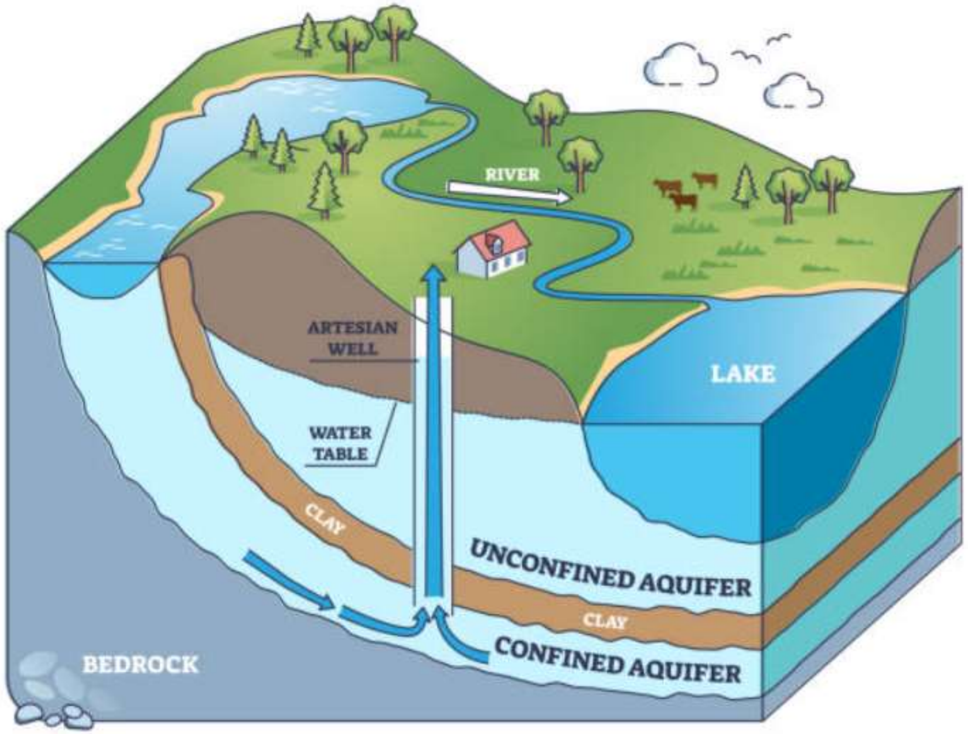
Surface Water: Water that is open to the atmosphere and results from overland flow. It is also said to result from surface runoff.

Source of surface water: *Streams, rivers, ocean, lakes, ponds, rain catchments, etc.*

Surface Water Source

Groundwater: Water that is below the earth's crust. Water between the Earth's crust and the 2,500-foot level is considered usable freshwater.

Source of groundwater: *Wells, aquifers, springs, etc.*



Aquifer: Permeable rocks have tiny spaces between the solid rock particles that allow water to pass through and be held within the rock structure.

Spring: A surface feature where, without the help of man, water issues from rock or soil onto the land.

Ground Water Source

Water Pollution

It is the contamination of water bodies with harmful substances, such as toxic chemicals, microorganisms, or waste, thereby making them unsafe for human use, animals, plants, and aquatic life.

Types of Water Pollution

- *Surface Water Pollution*
- *Groundwater Pollution*
- *Chemical Pollution*
- *Microbiological Pollution*
- *Nutrient Pollution*
- *Suspended Matter Pollution*
- *Oxygen-depleting Pollution*

Surface Water Pollution

Surface water pollution affects visible bodies of water such as lakes, rivers, and oceans. This pollution results from both point sources (*direct discharge from factories and sewage treatment plants*) and nonpoint sources (*runoff from urban areas and agriculture*).

Sources: Industrial effluents, untreated wastewater, agricultural runoff, litter, and oil spills.

Impact: Surface water pollution degrades water quality, harms aquatic life, and can render the water unsafe for human use, including drinking, fishing, and recreational activities.

Groundwater Pollution

Groundwater is polluted when harmful substances seep into the ground and contaminate underground water reserves, and it is difficult to clean due to slow natural filtration and long residence times.

Sources: Agricultural runoff containing pesticides and fertilizers, leakage from septic tanks and sewage systems, chemical spills, landfills, oil, etc.

Impact: Pollutants such as nitrates, heavy metals, and pathogens make groundwater unsafe for drinking and agriculture.

Note: Groundwater pollution is difficult to detect and clean, making it a long-term issue for affected communities.

Chemical Pollution

One of the most widespread forms of water pollution, chemical contamination occurs when hazardous chemicals are released into water bodies. This type of pollution can have both immediate and long-term toxic effects on aquatic life and human health.

Sources: Industrial discharges, agricultural runoff containing pesticides and fertilizers, mining operations, and improper disposal of household chemicals.

Impact: These pollutants can accumulate in the water supply, causing severe health problems such as cancer, neurological damage, and reproductive issues, while also poisoning aquatic life.

Microbiological Pollution

Microbiological pollution occurs when harmful microorganisms, such as bacteria, viruses, and protozoa, are present in water. These pathogens can cause diseases when water is consumed, used for bathing, or used for irrigation.

Sources: Human and animal waste, agricultural runoff, untreated sewage, and stormwater.

Impact: This type of contamination is especially common in areas with inadequate water treatment facilities, leading to outbreaks of waterborne diseases such as cholera, dysentery, and typhoid.

Nutrient Pollution

Excessive nutrients, such as nitrogen, magnesium, phosphorus, etc., in water bodies can stimulate the overgrowth of algae, leading to imbalances in aquatic ecosystems.

Sources: Fertilizers from agricultural runoff, wastewater, detergents, and livestock waste.

Impact: Algal blooms, which result from nutrient overload, block sunlight and deplete oxygen in the water, killing fish and other aquatic organisms. Additionally, some algae release toxins that can further degrade water quality and pose health risks to humans and animals.

Suspended Matter Pollution

Suspended matter refers to pollutants that do not dissolve in water but remain suspended as solids or particles. These materials can be either organic or inorganic and have a range of adverse effects on water quality.

Sources: Erosion from land, runoff from agricultural fields, industrial waste discharges, and algae blooms.

Impact: Suspended particles can settle on the waterbed, disrupting aquatic habitats by smothering organisms. Floating particles block sunlight and reduce oxygen levels, impairing photosynthesis in aquatic plants and threatening fish and other species.

Oxygen Depletion Pollution

Oxygen depletion occurs when excessive organic matter, such as dead algae or plant matter, decomposes in water, consuming large amounts of dissolved oxygen.

Sources: Agricultural runoff containing fertilizers, wastewater containing organic materials, and algal blooms caused by nutrient pollution.

Impact: Oxygen depletion creates "*dead zones*" where most aquatic life cannot survive, significantly altering the ecosystem. Additionally, certain species of algae release toxins that can contaminate water supplies, causing health problems for humans and animals.

Lecture No. 2

- Sources of Contamination
- Pollutants, Their Types, and Effects

Sources of Contamination

Sources of water contamination can be categorized broadly into three types:

i. ***Point source***

ii. ***Non-point source***

iii. ***Transboundary source***

1. Point Source

It is a **single identifiable source of pollution** where pollutants are directly discharged into a water body. Hence, easier monitoring and tracking of the source, as well as convenient regulation, are possible, as contamination can be traced back to a specific origin.

Examples:

- **Industrial Discharges**
- **Sewage Treatment Plants**
- **Animal Factory Farms**

Industrial Discharge

2. Non-point Source

Nonpoint source pollution **does not originate from a single source** but results from pollutants being washed into water bodies over a large area. This type is more difficult to control because no leading or traceable source can be identified.

Examples:

- **Agricultural Runoff**
- **Urban Runoff**
- **Abandoned Mines**
- **Atmospheric Deposition**

Urban Runoff

3. Transboundary Source

Transboundary pollution occurs **when polluted water from one country flows into the water of another**. It can involve large and sudden disasters or long-term, chronic industrial or agricultural runoff.

Oil Spills: The oil spill in one country's territorial waters would easily spread and contaminate the waters of other countries. An oil slick can travel hundreds of miles, impacting marine ecosystems and coastal communities.

Downriver Contamination: Industry or farm pollutants upstream in the current can carry downstream into other regions or countries. International boundary-crossing rivers may carry industrial wastes, agricultural runoff, or untreated sewage across borders into other countries.

Oil Spills in the Ocean

Pollutants: These are substances introduced into the environment that have an adverse effect on living beings.

Types of pollutants

- Organic pollutants
  - ***Oxygen-demanding waste***
  - ***Synthetic organic compounds***
  - ***Oils***
- Inorganic pollutants
- Pathogens
- Nutrients
- Suspended solids and sediments
- Thermal pollutants
- Radioactive pollutants

1. Organic Pollutants

a) Oxygen-Demanding Waste:

- The wastewaters, such as domestic and municipal sewage, wastewater from food processing industries, canning industries, paper and pulp mills, breweries, distilleries, etc., have a considerable concentration of biodegradable organic compounds. These wastes undergo degradation and decomposition by bacterial activity.
- Dissolved oxygen (DO) in the water body will be consumed during the aerobic oxidation of organic matter in the wastewater. Hence, depletion of the DO will be a serious problem, adversely affecting aquatic life, if the DO falls below 4.0 mg/L. This decrease in DO is an index of pollution.

Food Waste

Waste Discharge from Breweries

b) Synthetic Organic Compounds: These are also likely to enter the ecosystem through various manmade activities such as the production of these compounds, spillage during transportation, and their uses in different applications. These include *synthetic pesticides*, *synthetic detergents*, *food additives*, *pharmaceuticals*, *insecticides*, *paints*, *synthetic fibers*, *plastics*, *solvents*, and *volatile organic compounds*. Most of these compounds are toxic and bio-refractory, meaning they are resistant to microbial degradation.

c) Oil: Oil enters water through oil spills, pipeline leaks, and wastewater from production and refining facilities. Being lighter than water, it spreads to the water surface, separating the water-air interface and thereby reducing DO levels.

Pesticides

Synthetic Waste (*Plastics*) in the River

2. Inorganic Pollutants

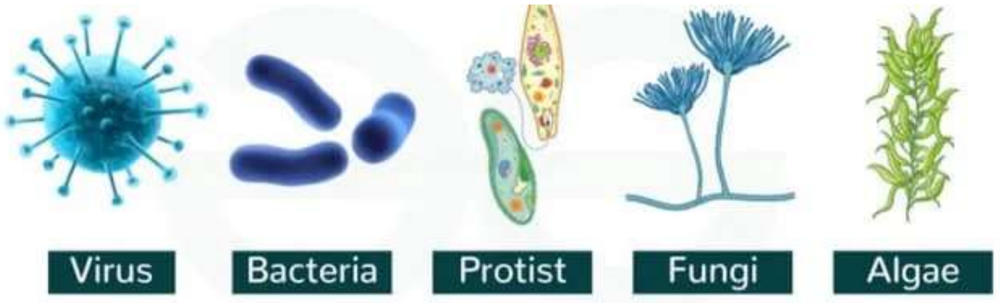
Through sewage and industrial waste, high concentrations of heavy metals and other inorganic pollutants are discharged into the water. **These compounds are non-biodegradable and persist in the environment for a longer time.** These pollutants include mineral acids, inorganic salts, metal compounds, cyanides, sulphates, etc.

- The accumulation of heavy metals may have an adverse effect on aquatic flora and fauna and may constitute a public health problem when used for food.
- Algal growth due to nitrogen and phosphorus compounds can be observed.
- Metals in high concentration can be toxic to biota, e.g., Hg, Cu, Cd, Pb, As, and Se.

Arsenic Poisoning

3. Pathogens

Pathogenic microorganisms enter water bodies through sewage discharge, a major source, or through wastewater from animal-based industries such as poultry farms. Viruses and bacteria can cause water-borne diseases, such as cholera, typhoid, dysentery, and infectious hepatitis in humans.

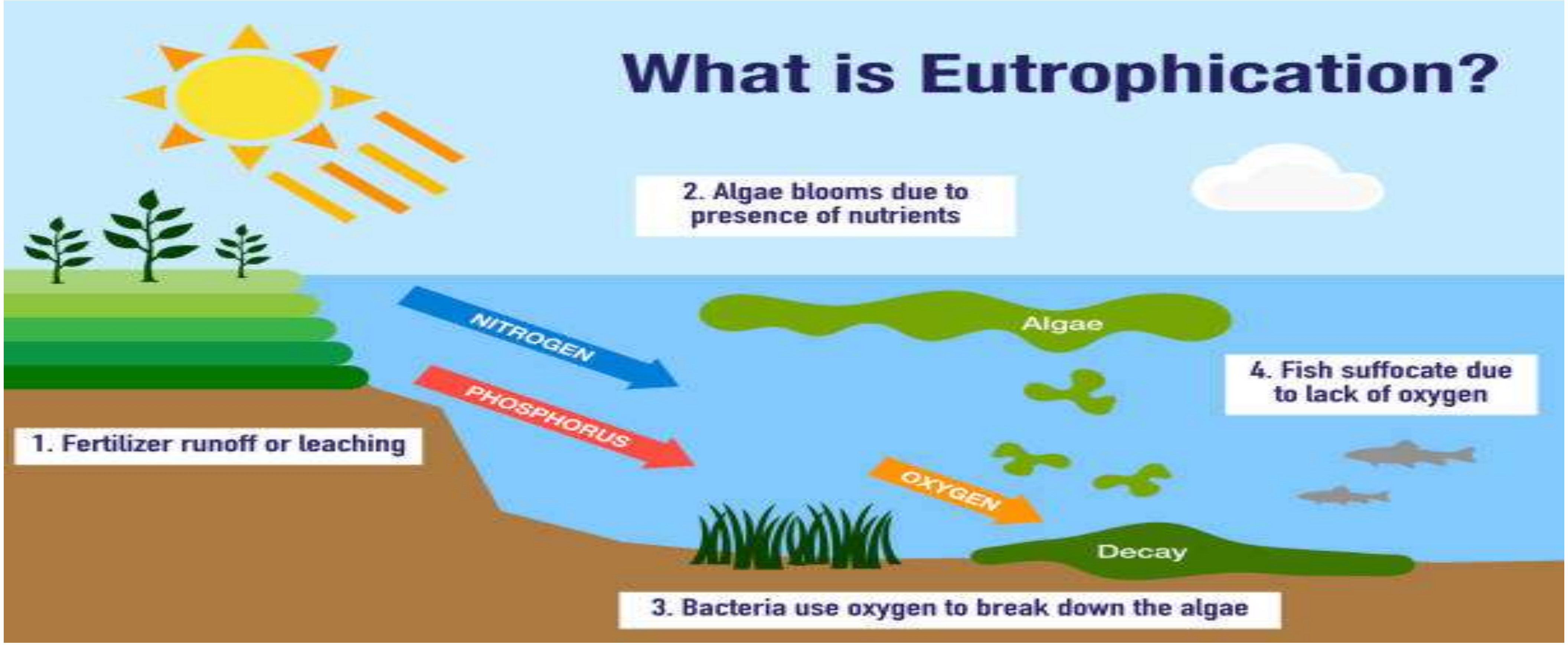


Pathogens

4. Nutrients

Agricultural runoff, wastewater from the fertilizer industry, and sewage contain substantial concentrations of nutrients, such as nitrogen and phosphorus. These waters supply nutrients to the plants and may stimulate the growth of algae and other aquatic weeds in receiving waters. Thus, the DO value of the water body is degraded. In the long run, a water body can reduce DO, lead to eutrophication, and eventually become a dead pool.

Eutrophication is a process in freshwater bodies where water receives excess inorganic nutrients, especially phosphorus, which stimulates the excessive growth of plants and algae.



Process of Eutrophication

Effects caused by eutrophication are as follows:

- Species diversity decreases
- Plant and animal biomass increase
- Turbidity increases
- Rate of sedimentation increases, causing shortening of the lifespan of the lake
- People swimming in eutrophic waters can have skin and eye irritation, gastroenteritis, and vomiting.
- High nitrogen levels in the water supply cause a potential risk, especially to infants under six months.

Algae Bloom

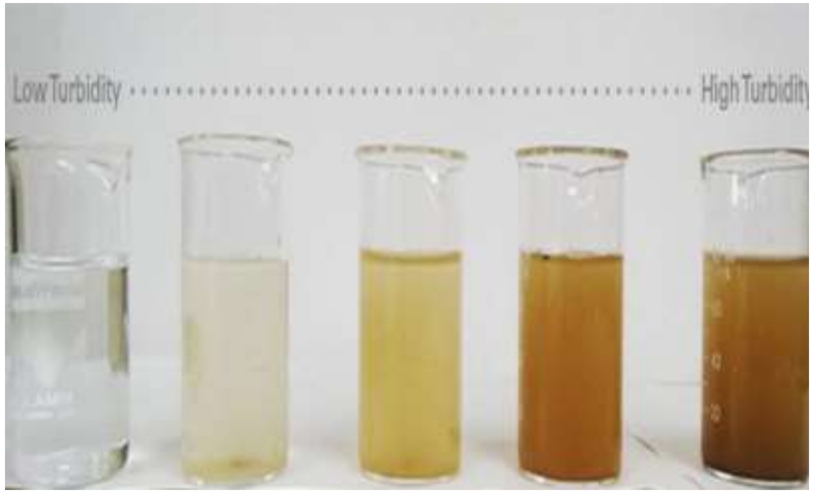
Blue Baby Syndrome or Cyanosis

5. Suspended Solids and Sediments

These comprise silt, sand, and minerals eroded from the land. These appear in the water through surface runoff during the rainy season and through municipal sewage systems.

- It leads to siltation, reducing the storage capacities of reservoirs.
- Presence of suspended solids can block the sunlight penetration in the water, which is required for photosynthesis by bottom vegetation.
- Deposition of the solids in the quiescent stretches of the stream or ocean bottom can impair the normal aquatic life and affect the diversity of the aquatic ecosystem.
- Finer suspended solids, such as silt and coal dust, may injure the gills of fish and cause asphyxiation.

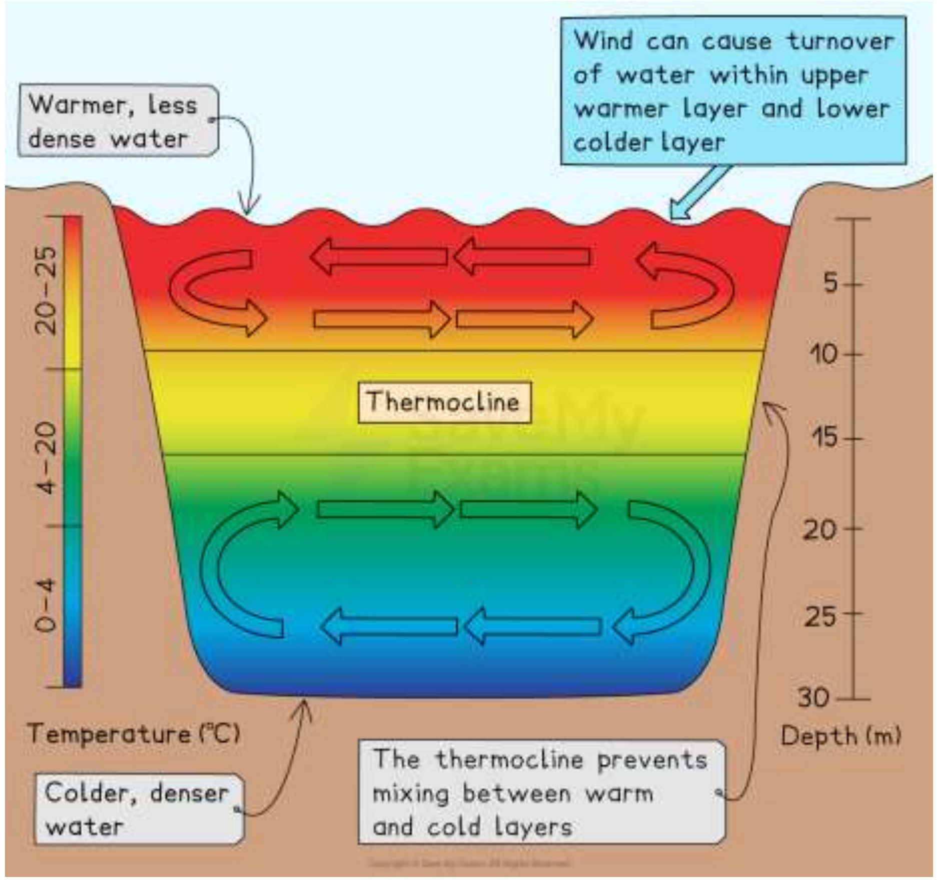
Suspended Matter in the Water Body



6. Thermal Pollutants

It results from the discharge of hot water from thermal and nuclear power plants and from industrial facilities that use water as a coolant. As a result of hot-water discharge, the water body's temperature increases.

- Rise in temperature reduces the DO content of the water, affecting the aquatic life adversely.
- Alters the spectrum of organisms, which can adapt to live at that temperature and DO level.
- When organic matter is also present, the bacterial action increases due to a rise in temperature; hence, resulting in a rapid decrease of DO.
- Discharge of hot water leads to **thermal stratification** in the water body, where hot water will remain on the top surface.



Thermal Stratification

7. Radioactive Pollutants

It is one of the most hazardous materials to handle from a safety perspective, as radioactive residue can persist for thousands of years. It includes Uranium mining, nuclear power plants, reactors, nuclear weapons, and medical research, among others.

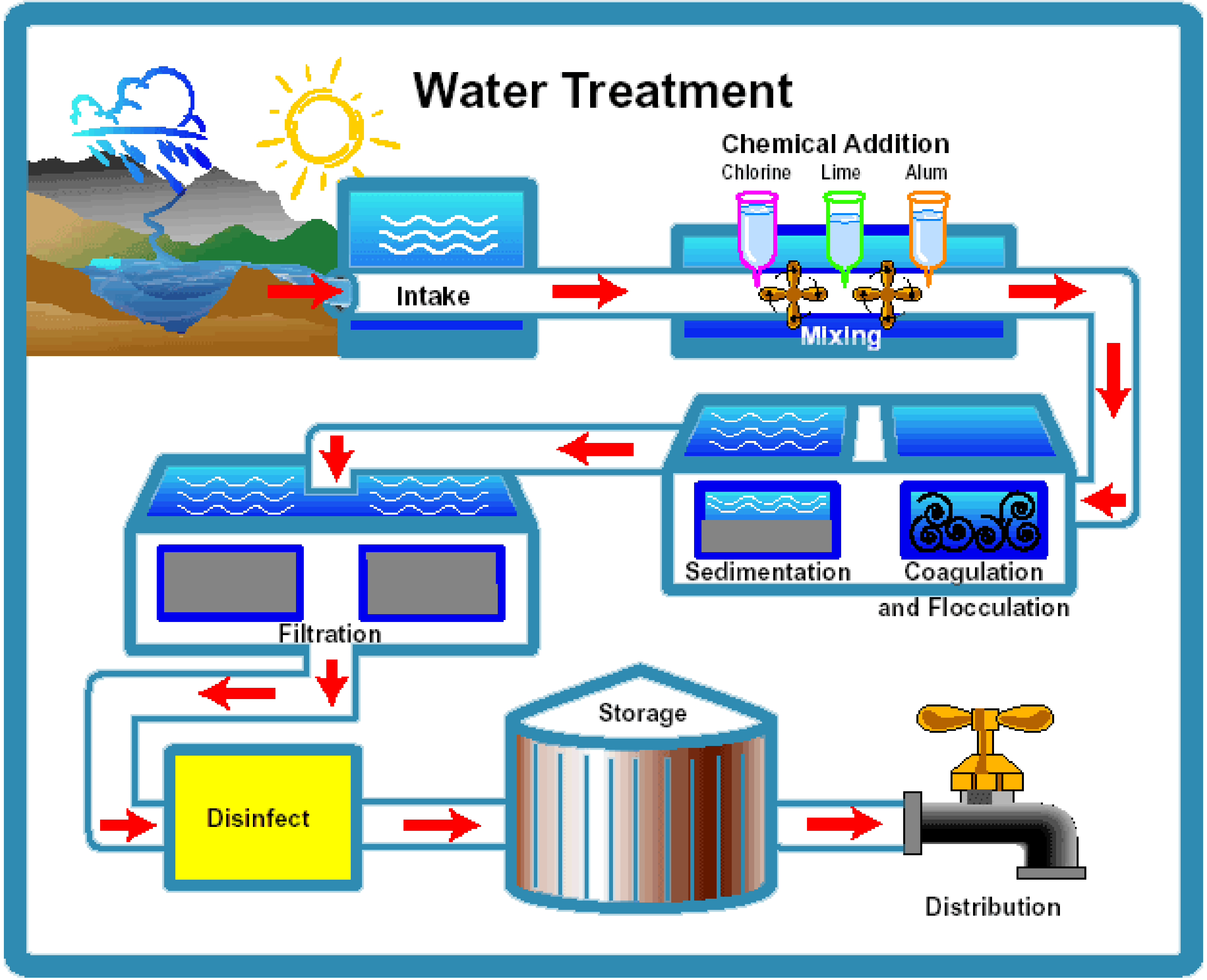
Persistent Long-term pollution of both groundwater and surface water due to radioactive material results in health impacts, including, but not limited to, cancer and genetic changes.

Note: The safe concentration for lifetime consumption is $1 \times 10^{(-7)}$ microcuries per milliliter.

Radioactive Discharge

Lecture No. 3

- **Municipal Water Treatment**
- **Disinfection Methods**



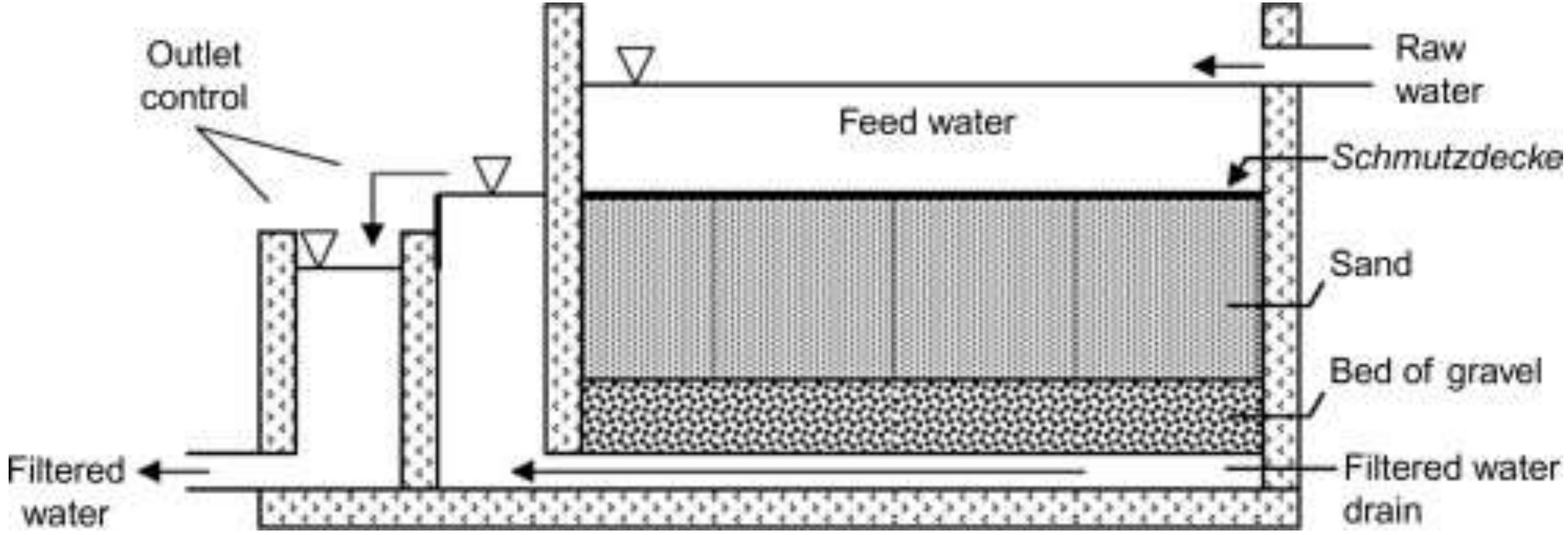
- **Coagulation and flocculation** are critical, sequential water treatment processes used to **remove suspended solids, colloids, and organic matter**.

- Coagulation involves **adding chemicals to neutralize** negative charges on particles. e.g., Aluminum sulfate (Alum), Aluminum Chlorohydrate, Ferric chloride, etc.

- Flocculation uses **gentle mixing to encourage these particles to clump** into larger, settleable flocs.

Municipal Water Treatment

1. **Slow Sand Filter -** It operates at a slower rate and relies on biological processes for filtration. The filtration rate is typically 0.1–0.3 m^3/h per m^2 of filter area. The process relies on biological activity in the upper layer of sand (***schmutzdecke***) to remove contaminants. This layer helps to trap and remove suspended solids, pathogens, and other impurities. No chemicals are required; removal is mainly physical and biological.



Working of Slow Sand Filter

Working Principle:

The water from sedimentation tanks enters the slow sand filter through a raw water inlet. This water is uniformly spread over a sand bed without disturbing it. The water passes through the filter media at an average rate of 2.4 to 3.6 $m^3/m^2/day$.

Then, raw water is allowed to stand and slowly percolate through the sand bed, and a thin biological layer called a ***schmutzdecke*** forms on the top few centimeters of the sand. It consists of bacteria, algae, protozoa, and microorganisms.

Microorganisms in the *schmutzdecke* consume and destroy pathogenic bacteria, viruses, and organic matter. Pathogens are removed by predation, natural die-off, and biochemical oxidation

Fine sand particles physically trap suspended solids, turbidity, and microorganisms as water percolates downward. Larger particles are retained near the surface, while finer particles are trapped deeper in the sand bed. At last, purified water is collected through under-drainage systems at the bottom of the filter

- During filtration, as the filter media becomes clogged due to impurities accumulating in the pores, the resistance to water passage and the head-loss also increase. When the head-loss reaches 60 cm, filtration is stopped.
- Approximately 2 to 3 cm from the top of the bed is scraped and replaced with clean sand before the filter is returned to service. The scraped sand is washed with water, dried, and stored for reuse in the filter during the next washing. The filter can run for 6 to 8 weeks before the sand layer needs to be replaced.

Suitability:

- Best for rural or small communities with low turbidity water.
- Suitable where land is available and water demand is moderate.
- Ideal for treating surface water with low pollution and low suspended solids.
- Used in municipal water treatment in developing regions.

Advantages	Limitations
• Simple design, operation, and maintenance requirements • Minimal power and chemical requirements • Low operational costs • Minimal sludge handling problems • Close operator supervision is not necessary • Low operator skill level • Biological process removes pathogens and other contaminants.	• Requires a large area • Labor-intensive manual filter cleaning • Use is limited to low turbidity water • The biological process can be impaired at low temperature or low nutrient content • Operation complexity increases when ozone pretreatment and Granular Activated Carbon (GAC) GAC are included.

2. **Rapid Sand Filter** – It is a key component in municipal water treatment, using layers of sand and gravel to physically filter out suspended particles at high flow rates, typically after chemical pre-treatment (coagulation/flocculation) and before disinfection. It requires frequent cleaning (*backwashing*) to maintain efficiency for large volumes of drinking water.

It is mainly of two types:

- Open Rapid Sand Filter or Gravity Filter
- Closed Rapid Sand Filter or Pressure Filter

ASSIGNMENT

Methods for Water Disinfection

Disinfection methods of water are based on their principle of action:

- **Physical or non-chemical methods:** In this method, disinfection occurs through the influence of physical factors (boiling, ultraviolet, electrolysis, reverse osmosis).
- **Chemical or reagent methods:** In this certain reagents (chlorine, ozone, non-oxidizing agents) are added to the water.
- **Combined methods:** It involves the combination of both technologies, such as ultrafiltration and chlorination.

Physical Methods of Water Disinfection

1. Boiling: By boiling, most pathogenic organisms are killed, making this one of the simplest and most accessible water sterilization methods.

Note: For full disinfection, water should be boiled for at least 5-10 minutes.

Advantages

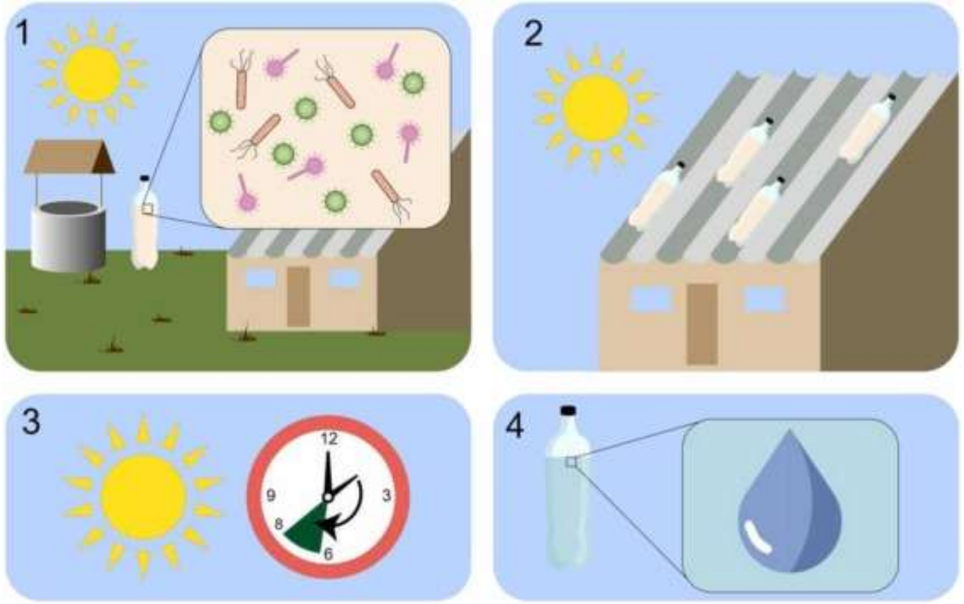
- Simplicity of execution
- No need for additional equipment
- Effectiveness against most pathogenic microorganisms
- In addition to disinfection, it reduces water hardness and turbidity

Disadvantages

- Significant increase in energy consumption with increasing water volume
- High duration
- Possibility of secondary contamination

2. Ultraviolet: Ultraviolet radiation, which is one of the components of the UV spectrum, is **able to break down the structure of thymine in the DNA of microorganisms.** As a result, bacteria and viruses lose their ability to reproduce in both water and the human body.

- The simplest method of UV water disinfection is the ***Solar Disinfection (SODIS) method***. Water, filtered to remove large mechanical particles larger than 50 microns, is poured into PET (*polyethylene terephthalate*) bottles, which are placed on a surface exposed to direct sunlight.



SODIS Method

- The main disadvantage of the SODIS method is that it needs active sunlight. The method is most effective in the strip between 35 degrees north and south latitude, where disinfection takes about six hours. As sunlight intensity decreases, the disinfection duration increases.

Advantages

- Easy-to-use method
- Does not require bulky equipment
- No need for constant dosing of reagents
- Does not introduce secondary pollution into the water, unlike disinfection with a reagent
- Low energy consumption.

Disadvantages

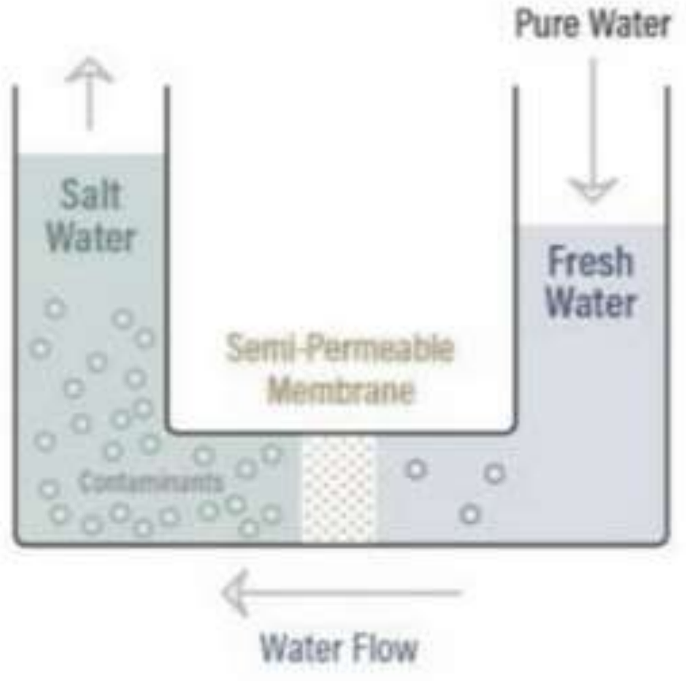
- Not effective against a wide range of microorganisms
- Regular replacement of the irradiation source is necessary
- Water must be free of mechanical particles, as they can reduce the method's effectiveness by 50%
- A long waiting time.

3. Reverse Osmosis (RO):

Osmosis is a naturally occurring phenomenon. It is the *movement of solvent or water through a semipermeable membrane, in which a weaker saline (low solute) solution tends to migrate toward a stronger saline (high solute) solution*.

For example, plant roots absorb water from the soil, or our kidneys absorb water from our blood.

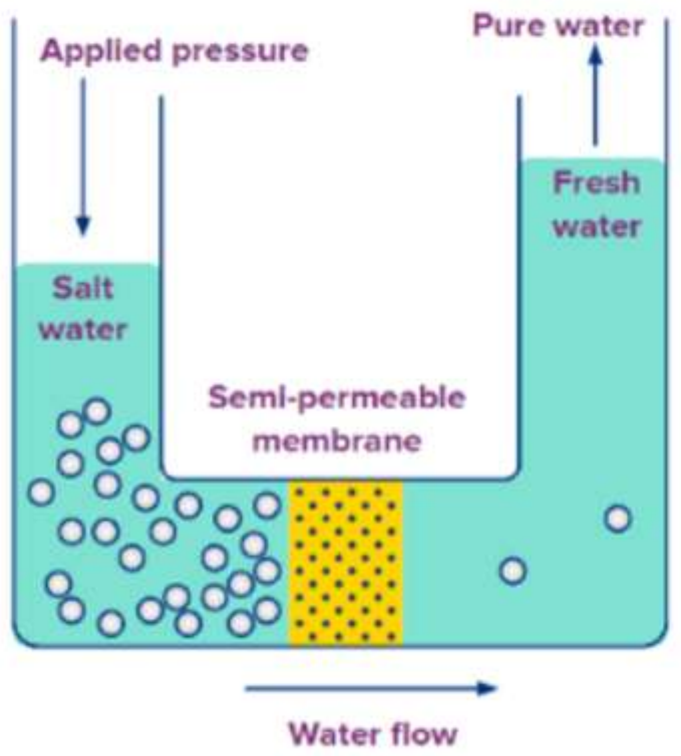
A container full of water with low salt concentration and another container full of water with high salt concentration, separated by a semipermeable membrane, the water with the **lower salt concentration would start to move towards the container with the higher salt concentration**.



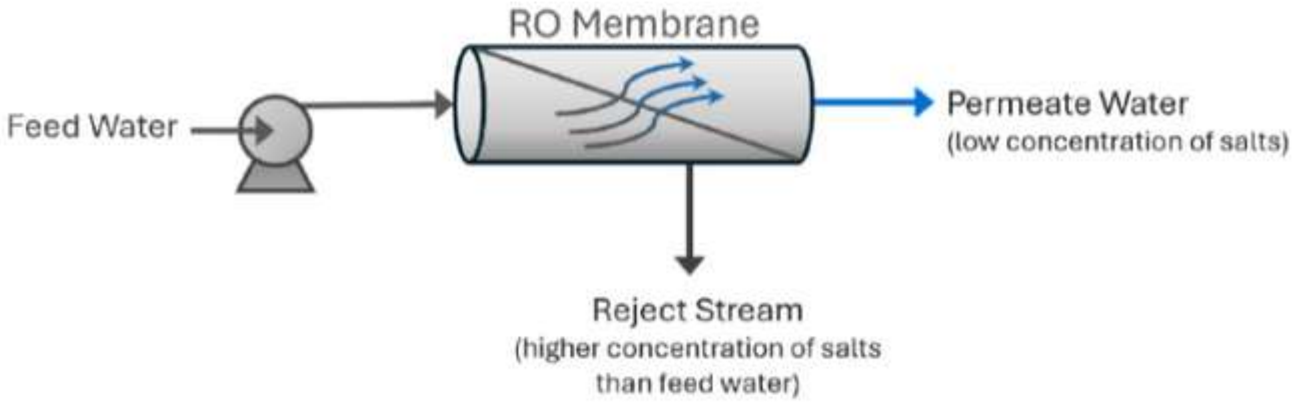
Osmosis Process

- A reverse osmosis membrane is a semi-permeable membrane that allows the passage of water molecules but not most dissolved salts, organics, bacteria, or pathogens.

- Osmosis occurs naturally without an external energy source, but reversing the process requires energy input. The water must be pushed through the RO membrane by applying pressure greater than the naturally occurring osmotic pressure.



Reverse Osmosis



RO system sizes are based on permeate flow. For example, a 100-gallon-per-minute (gpm) RO system will produce 100 gpm of permeate water

Advantages

- Highly effective at contaminant removal (approx. 90-95%):

Heavy metals such as lead, mercury, and arsenic.
Dissolved solids, including salts from water and nitrates.
Microorganisms such as bacteria, viruses, and parasites.
Chemical impurities, including fluoride, chlorine, and pesticides.

- Improves taste & odour
- Reduces sodium & hard water issues
- Energy efficient
- Versatility and wide applications

Disadvantages

- Waste water generation
- Removes some good Minerals (calcium, magnesium)
- Slower Process
- High initial cost & maintenance

Chemical or Reagent Methods

Reagent disinfection methods involve adding specific substances to the water. These compounds are divided into two groups:

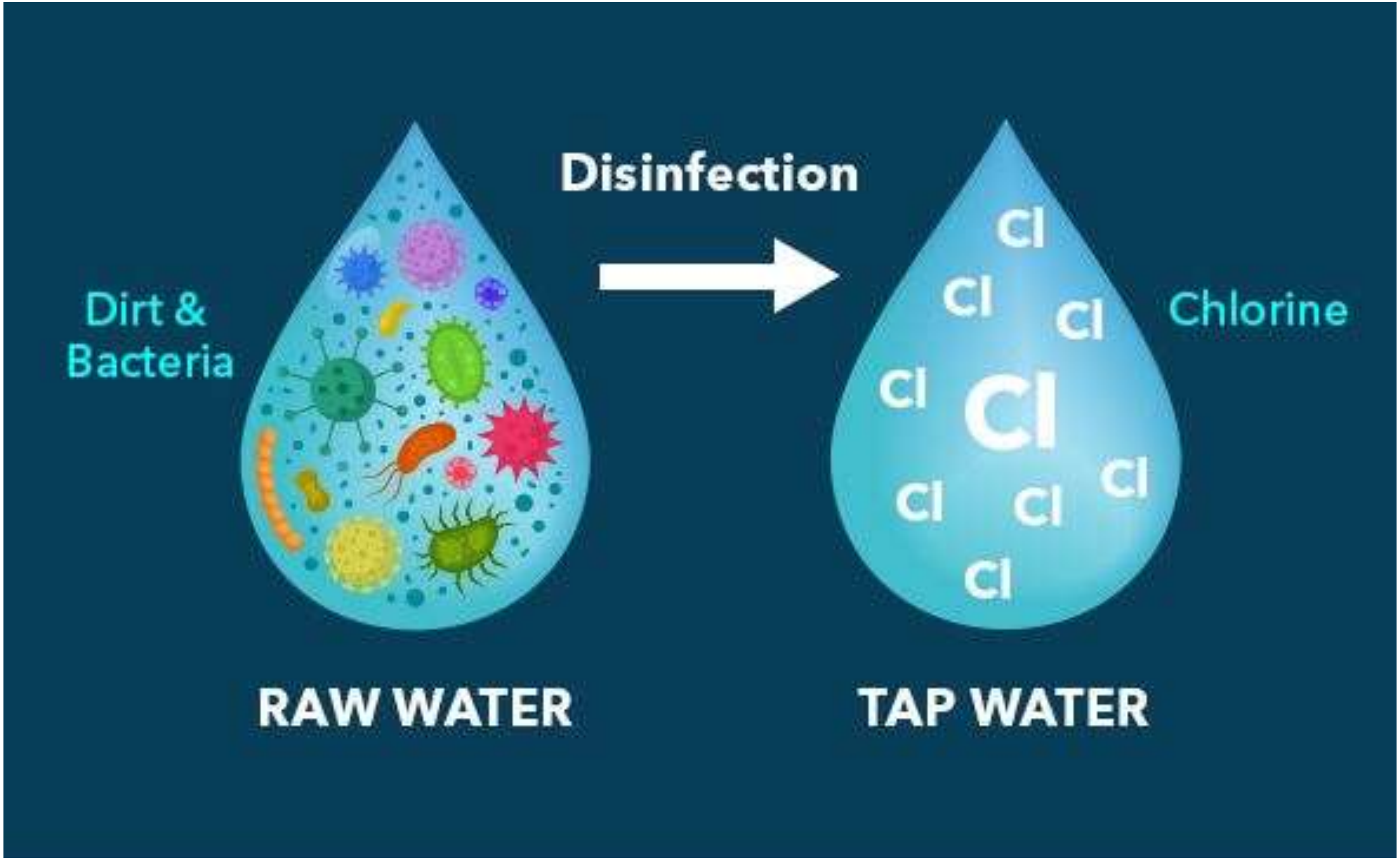
- **Oxidizing agents:** They *destroy the cell structure of microorganisms by oxidizing them*, while being reduced to less active compounds. Chlorination, Ozonation, etc.
- **Non-oxidizing agents:** They *provide bactericidal action through specific effects on microorganisms*, stopping their reproduction.

1. Chlorination

It involves adding compounds containing active chlorine to water, which can oxidize microorganisms and organic matter.

Main chlorine disinfectants:

- **Calcium hypochlorite (*Bleaching powder*)** – It is the most common disinfectant, and it is produced as granules that dissolve in water and are dosed in liquid form. These are not effective against cysts, and their effectiveness decreases with prolonged storage.
- **Chloroisocyanuric acid salts -** These are mainly used for **pools, tanks, secondary water supply systems**, but sometimes also for disinfecting drinking water in field conditions.
- **Chloramines –** These are used at **centralized water treatment plants** and are dosed into water in solution form. The advantage of this method is its long-lasting effect. Disadvantages include a stronger odor and lower effectiveness.
- **Chlorine dioxide** is one of the strongest chlorine oxidants, so it is not widely used in water treatment.
- **Chlorinated lime** (*a mixture of calcium hypochlorite, chloride, and hydroxide*).


Disinfection
Dirt &
Bacteria
Chlorine
Cl
RAW WATER
TAP WATER

Advantages

- Kills harmful and pathogenic microorganisms.
- Prevents waterborne diseases.
- Maintains water quality during distribution.
- Provides residual protection.
- Cost-effective water treatment method
- Easily available and economical

Disadvantages

- Unpleasant taste and odour.
- Formation of potentially harmful disinfection byproducts.
- Allergic reactions in some individuals.
- Corrosive to certain materials and equipment.
- Limited effectiveness against some parasites and viruses
- Safety concerns needed for storage and transportation

2. Ozonation

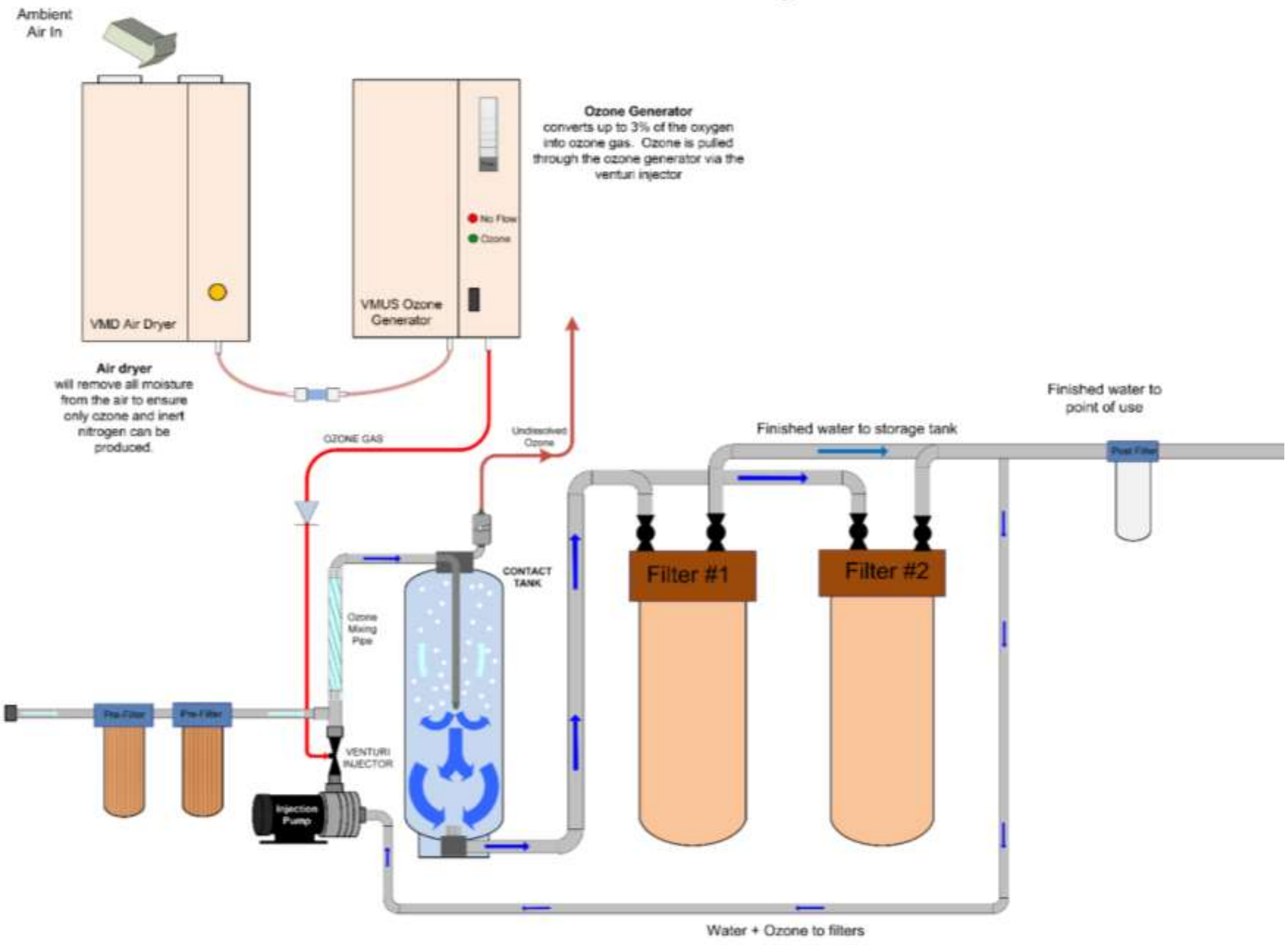
Ozone water treatment is an advanced method of purifying water using ozone (O_3), a powerful oxidizing agent. This process involves introducing ozone gas into water, where it quickly dissolves and reacts with contaminants.

- Ozone is highly effective at destroying bacteria, viruses, and other microorganisms, making it an excellent choice for disinfection in water treatment systems.

- As an oxidant, ozone can react with various substances by accepting their electrons, which allows it to break down complex organic compounds and inactivate pathogens.

- Ozone leaves no chemical residues as it quickly decomposes back into oxygen.

Ozone Filtration System



Ozone water treatment working

Ozone treatment for water purification is a multi-step process that harnesses the powerful oxidizing properties of ozone gas. The process begins with ozone generation, typically using an ozonator for water treatment. This device creates ozone by exposing oxygen molecules to an electrical discharge, splitting them and allowing them to recombine into ozone (O_3).

Ozone is then introduced into the water using various methods, such as bubble diffusion or injection systems. Once in the water, ozone rapidly dissolves and begins to react with contaminants. Its strong oxidizing properties allow it to break down organic compounds, inactivate microorganisms, and oxidize metals such as iron and manganese. As ozone encounters these substances, it transfers an oxygen atom to them, effectively destroying or altering their structure. Any remaining ozone quickly decomposes back into oxygen, leaving no chemical residues in the treated water.

Advantages

- High efficiency against a wide range of contaminants, including bacteria, viruses, and odor-causing compounds, resulting in clean, fresh-tasting water.
- Removes foreign tastes and odors.
- Environmentally friendly and chemical-free process
- Reacts rapidly with contaminants, providing fast and efficient water purification

Disadvantages

- Requires expensive equipment, high maintenance costs, etc.
- Increased safety requirements and personnel training
- Formation of secondary products that are toxic to humans in some cases.
- Unlike chlorine, ozone doesn't provide long-lasting disinfection in water distribution systems
- It has a complex working system and consumes a significant amount of energy

Non-Oxidising Agents

Non-oxidizing agents work differently from their oxidizing counterparts. These agents **interfere with critical biological functions inside the microorganism**, such as enzyme activity, protein synthesis, or cell membrane integrity.

The result is a more selective, controlled form of microbial elimination, ideal for systems where material compatibility and long-term stability matter.

Non-oxidizers are especially effective in targeting biofilm formation, which can insulate bacteria from oxidizing agents and severely impact system efficiency.

Common non-oxidizing biocides include **isothiazolinones, glutaraldehyde, DBNPA, and polyquats**.

Advantages

- Lower risk of material degradation
- Minimal production of toxic by-products
- Targeted action against specific microbial threats
- Better efficacy in biofilm and low-flow areas

Disadvantages

- Environmental persistence & toxicity
- Slow kill rates
- It can be more expensive and, in some cases, require precise, complex dosing

Environmental Sciences & Renewable Energy (ME-306)

<u>Module – 3</u>

- ***Brief introduction to noise pollution***
- ***Sources of noise pollution***
- ***Measurement and prevention of noise pollution***

Lecture No. 4

- **Introduction to Noise Pollution**
- **Sources of Noise Pollution**
- **Importance of Noise Measurement**
- **Measurement of Noise Level**
- **Control of Noise Pollution**

Noise Pollution

- Noise is derived from the Latin word – ***nausea (disgust or discomfort).***
- Sound is caused by the vibration in the transmitting medium that can be heard. When sound is loud and unpleasant, it is called noise. **All noise is sound**, but **not all sound is noise.** It is measured in **decibels (dB)**.
- Noise pollution is considered to be the **presence of any unpleasant or disturbing sound that affects the physiological and psychological health and well-being of humans, wildlife,** and other living beings.
- The WHO (*World Health Organization*) has set an optimal noise level of **45 dB during the day** and **35 dB at night**.
- *A decibel value above 80 is considered to be noise pollution*.

Sources of Noise Pollution

Transportation: Road traffic, railways, aircraft, etc.

Industrial activities: Factories, generators, heavy machinery, etc.

Construction: Drilling, hammering, demolition, etc.

Social & domestic sources: Loudspeakers, music systems, household appliances, etc.

Urban activities: Markets, festivals, public events, etc.

Other Sources: Human shout, crowd, etc.

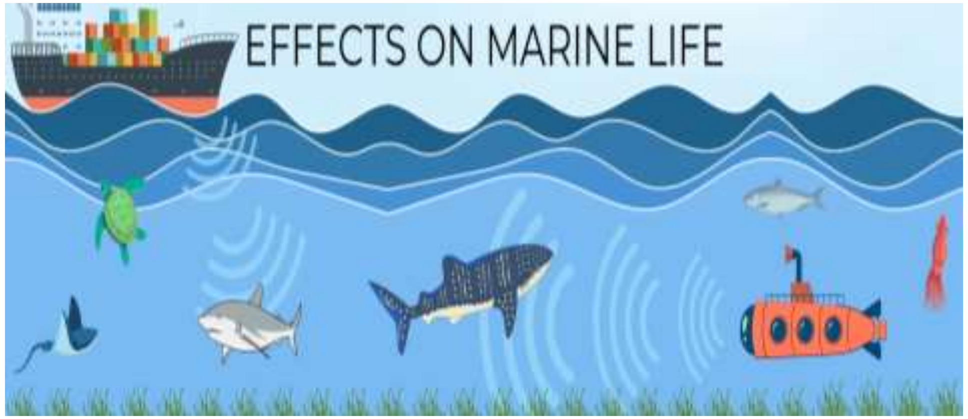
Sources of Noise Pollution

Effects of Noise Pollution

- Hearing loss and tinnitus
- Sleep disturbance and fatigue
- Stress, anxiety, irritation, etc.
- Reduced concentration and productivity
- Increased risk of hypertension and heart disease
- Alters migration and feeding patterns
- Causes habitat abandonment
- Disturbs ecological balance
- Reduces overall quality of life in urban areas
- Can damage the building and structures

Tinnitus

Note - Approximately 63 million people in India suffer from significant auditory impairment



Importance of Noise Measurements

- **To assess the impact of noise on human health:** Measuring noise levels can help to determine the potential health risks associated with exposure to noise and to develop strategies to reduce those risks.

- **To comply with noise regulations:** Many countries and jurisdictions have noise regulations that set limits on the amount of noise. Measuring noise levels can help ensure these regulations are met and identify sources of excessive noise.

- **To design and evaluate noise control measures:** It identifies the sources of noise and designs and evaluates noise control measures. This can be important for reducing noise levels in workplaces, homes, and other environments.

- **To conduct research on noise:** The research can help to improve our understanding of the risks associated with noise and to develop new strategies for noise control.

- **To raise awareness of noise pollution:** Measuring noise levels can help to raise awareness of the issue of noise pollution and its potential health effects. This can encourage individuals and organizations to take steps to reduce noise levels and to protect their health.

Measurement of Sound

Sound is measured based on its **intensity, pressure, and frequency**. The most common unit used is the ***decibel* (dB)**.

*Note - **Deci** comes from the Latin word for **10**, and **bels** comes from the scientist's name, **Alfred Graham Bell.***

Sound Intensity: The amount of sound energy passing through a unit area per second. It's unit is W/m^2.

I =P/A ; *where* ***P = power, A = area***

- The range of sound intensity is very large; thus, a logarithmic scale (decibel scale) is used.
- A dB is basically 10 times the logarithm of the ratio of two sound intensities (I) or pressures (p).

in terms of sound intensity, $dB = 10 \log_{10} (I / I_0)$

in terms of sound pressure, $dB = 10 \log_{10} (p / p_0)$

Sound Intensity Level (L_i): It indicates how much sound energy per unit area is flowing through a surface. It is measured in dB.

$$L_i = 10 \log_{10} (I / 10^{-12})$$

Where, I ***= measured intensity,***

$I_0 = 10^{-12}\ W/m^2$ ***(reference quantity, i.e., threshold value of human ear at 1Hz)***

Q: Measured intensity (I) = $10^{-4}\ W/m^2$ ***then find*** L_i ***= ………***

Ans: $L_i = 10 \log_{10} (10^{-4} / 10^{-12}) = 10 \log_{10} (10^{8}) = 80dB$

Relation b/w sound intensity and pressure, $I_0 = (p_0)^2/\rho c$

Where,

$I_0 = 10^{-12}\ W/m^2$ *(standard reference quantity)*

ρ = density of air ≈ 1.2 kg/m³
c = speed of sound ≈ 343 m/s
p_0 = reference pressure value

After calculation get $p_0 = 20\ \mu Pa$

$$\log(I) = \log(p^2) = 2\log(p)$$

Sound Pressure Level (L_p): $L_p = 10 \log_{10} (p_{rms} / 20)^2$ **or** $20 \log_{10} (p_{rms} / 20)$

Where, p_{rms} = ***measured sound pressure (in µPa),*** $p_0 = 20\mu Pa = 2\times10^{-5} Pa$

- If two independent sound pressures p_1 and p_2 are given in Pa, then

$$p_{rms} = \sqrt{(p_1^{\ 2} + p_2^{\ 2})};\ L_p = 20 \log_{10} (p_{rms} / 2\times10^{-5})$$

Note *– Sound-measuring instruments measure the r.m.s. pressure; hence, the sound pressure is computed by squaring their r.m.s. value.*

- ***1dB is the lowest sound level perceived by the human ear, and the maximum of 180dB can be tolerated.***
- ***An increase of 10dB in sound intensity level will increase the sound intensity by 10 times.***
- ***An increase of 10dB in sound pressure level will increase the sound pressure (p_{rms}) by 3.16 times.***
- ***An increase of 20dB sound pressure level will increase the sound pressure by 10 times.***
- ***If the sound level increases by 10 dB, the sound is perceived as twice as loud.*** Perceived loudness is measured in ***sones.***

Addition of Sound Levels (in dB): Sound levels cannot be added directly because the decibel (dB) scale is logarithmic.

- If two sound levels in dB are L_1 and L_2, then

$$L_{total} = 10 \log_{10} (10^{L_1/10} + 10^{L_2/10})$$

- When **multiple sound intensity level measurements** (*N numbers*) are **given in dB**, use the **logarithmic addition method**:

$$L_{total} = 10 \log_{10} (10^{L_1/10} + 10^{L_2/10} + 10^{L_3/10} + \ldots)$$

- **Average sound level (L_{avg}) = $10 \log_{10} [(10^{L_1/10} + 10^{L_2/10} + 10^{L_3/10} + \ldots) / N]$**

Quantity	**Addition Method**
Pressure (Pa)	Root sum square; $p_{rms} = \sqrt{(p_1^2 + p_2^2)}$
Intensity (W/m²)	Direct addition; $I_{total} = I_1 + I_2$
Level (dB)	Logarithmic addition

For example, $L_1 = 60dB$, $L_2 = 70dB$, $L_3 = 65dB$ then

$$\mathbf{L_{total}} = 10\log_{10}(10^{60/10} + 10^{70/10} + 10^{65/10})$$

$$= 10\log_{10}(14162277.66) = \mathbf{\sim 71.50\ dB}\ \text{then,}$$

$$\mathbf{L_{avg}} = 10\log_{10}[(10^{60/10} + 10^{70/10} + 10^{65/10})/3] = \mathbf{\sim 66.74\ dB}$$

If ***N sources*** each produce the same sound level ***L,*** then

$$\mathbf{L_{total} = L + 10\log_{10}(N)}$$

For example, *two sound sources each produce a 50 dB sound level*

$$(L_1 = 50, L_2 = 50),\ \text{then}\ L_{total} = 50 + 10\log_{10}(2) = \sim 53dB$$

If there are

2 equal sources → L + 3 dB ; 3 equal sources → L + 4.8 dB
4 equal sources → L + 6 dB ; 10 equal sources → L + 10 dB

Frequency is the number of vibrations of a sound wave per second. **Unit – Hertz (Hz).**

$$f = 1/T$$

***f* = frequency (Hz), *T* = time period (seconds)**

- One Hz means one vibration per second.
- Human hearing range is *20 to 20kHz. But more sensitive to middle-range frequencies, i.e., 500 to 5000 Hz.* Hence, when measuring sound loudness, electronic filtering circuits (**weighting networks**) are used in sound-pressure-measuring meters.

Frequency Weighting Networks: The human ear does not respond equally to all frequencies. The weighting network filters out certain frequencies and permits the measuring meter to respond more to some frequencies than others.

Weighting	Use
dB(A)	Most commonly used; filters low < 200Hz and high > 10kHz frequecy
dB(B)	Rarely used
dB(C)	High-level industrial noise

Measurement of Noise Level

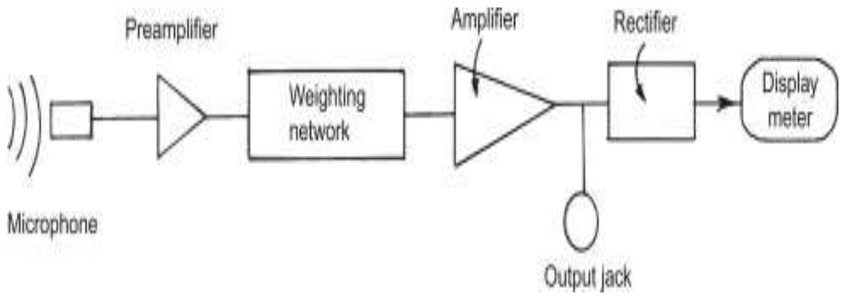
- Noise level is usually measured as Sound Pressure Level in decibels (dB).

Sound Level Meter (SLM): It comprises a microphone, a preamplifier, a frequency weighting network, and a display. The microphone converts the sound wave energy into an electrical signal.

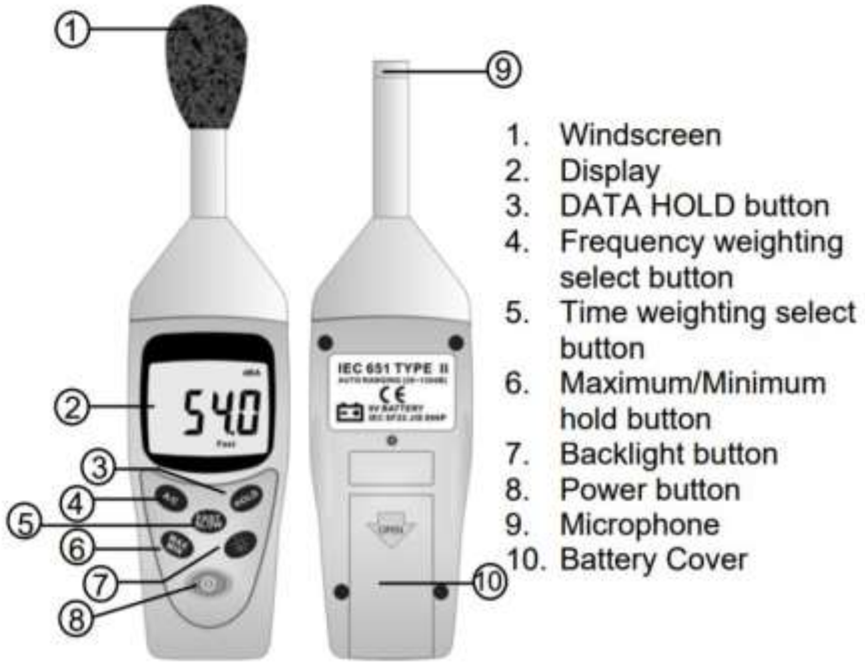
- The electrical signal from the microphone is at a very low level, so it is amplified by a preamplifier before it is processed by the main processor.

- The electrical signal is further modified by the weighting network, followed by further amplification through the amplifier.

- The rectifier converts the AC signal to DC, allowing the display meter to directly register the sound pressure level in decibels.



Block Diagram of Sound Level Meter



Sound Level Meter

Equivalent Continuous Noise Level (L_{eq}): Represents a constant noise level having the same energy as fluctuating noise over time (t).

$$L_{eq} = 10\log \sum_{i=1}^{n} (10)^{L_i/10} \times t_i$$

Here,

Note - $L_{eq(8)}$ means that the measurement is based on 8 hours, whereas L_{eq} represents for one hr measurement.

n = number of sound samples
L_i = noise level of i[th] sample
t_i = time duration of i[th] sample ***(fraction of total sample time)***

For example:

Q. The fluctuation noise level of 95 minutes is as follows:

80dB for 10 minutes, 60dB for 80 minutes, and 100dB for 5 minutes. Find L_{eq}

Ans – L_1 = 80 dB, t_1 = 10/95,
L_2 = 60 dB, t_2 = 80/95,
L_3 = 100 dB, t_3 = 5/95

After putting above in formula get, L_{eq} = 87.3dB

Level in decibels	**Everyday example**	**Times more intense**	**Times louder**
10dB	Rustling or falling leaves.	1	1
20dB	Watch ticking.	10	2
30dB	Birds flying by.	100	4
40dB	Quiet conversation.	1,000	8
50dB	Louder conversation.	10,000	16
60dB	Quiet traffic noise.	100,000	32
70dB+	Louder traffic	1 million	64
80dB+	Loud highway noise at close range	10 million	128
85dB	***Hearing damage after about 8 hours.***		
100dB	Jackhammer (pneumatic drill) at close range	1 billion	512
100dB	***Hearing damage after about 15 minutes.***		
110dB+	Jet engine at about 100m	10 billion	1024
120dB	***Threshold of pain. Hearing damage after very brief exposure.***		

Noise level standards in India

The GoI notified the rule under the **Environment Protection Act and the Noise Pollution (Regulation and Control) Rules 2000**, for the regulation and control of noise pollution.

Ambient Air Quality Standards in Respect of Noise (India)

Area code	Category of area/zone	Limits in dB(A) leq*	
		Day time	Night time
(A)	Industrial area	75	70
(B)	Commercial area	65	55
(C)	Residential area	55	45
(D)	Silence zones	50	40

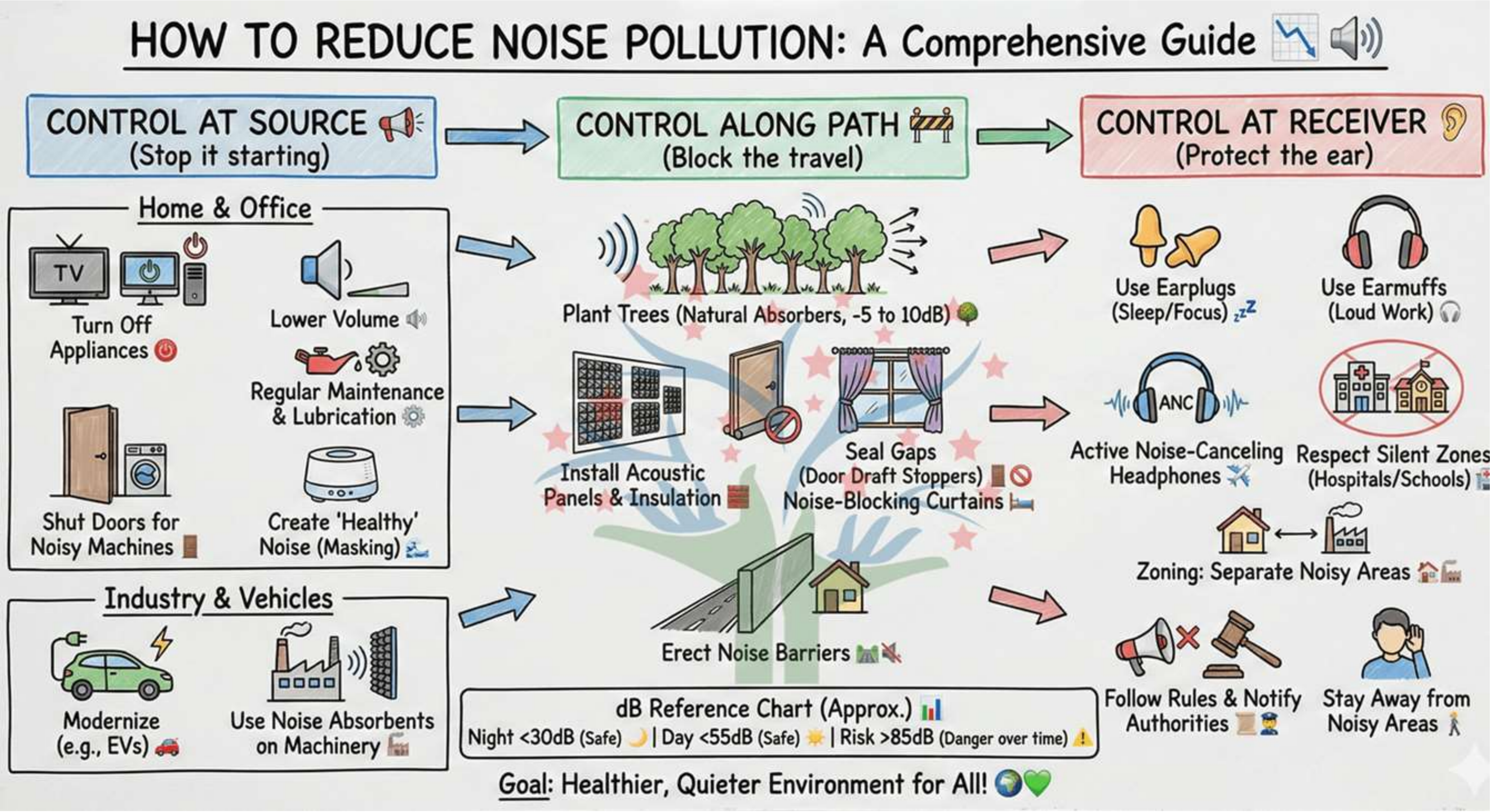
*dB(A) Leq denotes the time weighted average of the level of sound in decibels on scale A which is relatable to human hearing. Source: Central Pollution Control Board, India

Day-time *– 6 AM to 10 PM,* ***Night Time*** *– 10 PM to 6 AM*

Silence Zone *– It is the area comprising not less than 100 m around hospitals, educational institutions, courts, and areas declared by a competent authority.*

Noise Pollution Control

- **Enforcing noise pollution standards:** The Central Pollution Control Board (CPCB) is responsible for enforcing noise pollution standards in India. It needs to take strict actions to enforce noise pollution standards.
- **Educating the public about noise pollution:** This can be done through public awareness campaigns and educational programs in schools.
- **Using noise-reducing technologies:** There are a number of noise-reducing technologies available that can be used, e.g., soundproofing, noise barriers, tree plantation, silencers, etc.
- **Reduce noise at the source (*Enclosing noisy machinery, installing soundproofing, etc.*)**
- **Block noise transmission (*Using sound barriers, closing windows and doors, earplugs, etc.*)**
- **Proper town planning techniques (*proper segregation of residential and commercial zones*)**
- **Strict implementation of the Motor Vehicle Act (MVA)**


HOW TO REDUCE NOISE POLLUTION: A Comprehensive Guide
CONTROL AT SOURCE
(Stop it starting)
CONTROL ALONG PATH
(Block the travel)
CONTROL AT RECEIVER
(Protect the ear)
Home & Office
TV
Turn Off Appliances
Lower Volume
Regular Maintenance & Lubrication
Shut Doors for Noisy Machines
Create 'Healthy' Noise (Masking)
Industry & Vehicles
Modernize (e.g., EVs)
Use Noise Absorbents on Machinery
Plant Trees (Natural Absorbers, -5 to 10dB)
Install Acoustic Panels & Insulation
Seal Gaps (Door Draft Stoppers)
Noise-Blocking Curtains
Erect Noise Barriers
Use Earplugs (Sleep/Focus)
Use Earmuffs (Loud Work)
ANC
Active Noise-Canceling Headphones
Respect Silent Zones (Hospitals/Schools)
Zoning: Separate Noisy Areas
Follow Rules & Notify Authorities
Stay Away from Noisy Areas
dB Reference Chart (Approx.)
Night <30dB (Safe) | Day <55dB (Safe) | Risk >85dB (Danger over time)
Goal: Healthier, Quieter Environment for All!

Environmental Sciences & Renewable Energy (ME-306)

<u>Module – 3</u>

- ***Lithosphere and Soil Profile***
- ***Soil Contamination***
- ***Sources of Soil Contamination***
- ***Radioactive Pollution***

Lecture No. 5

- Lithosphere and Soil Profile
- Soil Contamination
- Radioactive Pollution

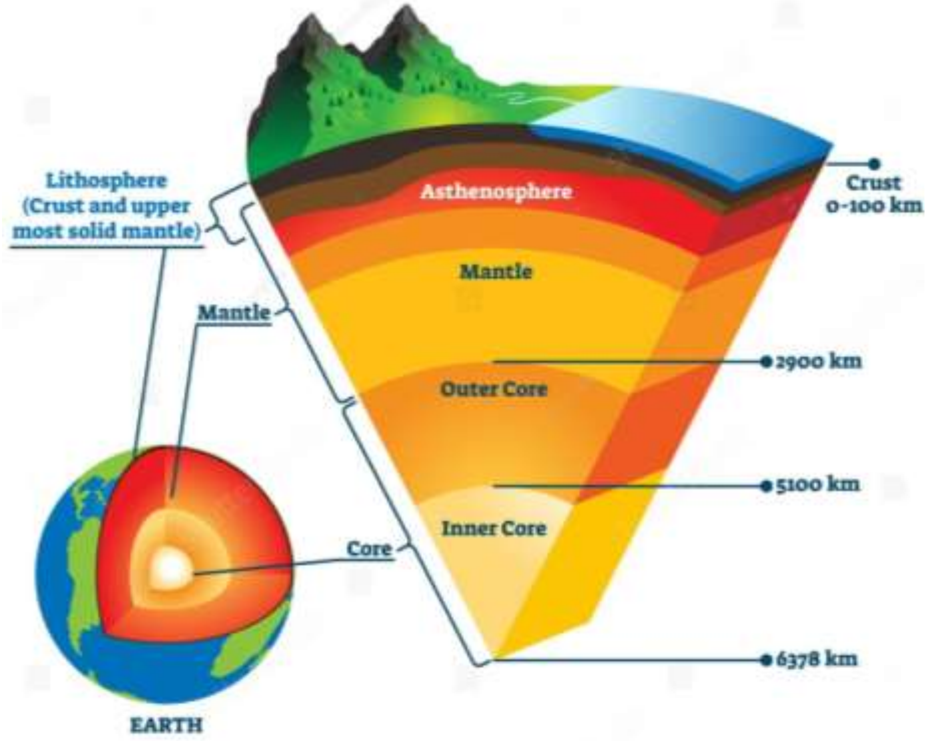
Structure and Composition of the Earth

- The Earth may be divided into three zones: **the crust, mantle, and the core.**

- The crust is the part of the Earth we live on and has less than 1% of the Earth's total mass and only about 0.5% of its radius. It appears to be solid.

- Mantle is the thick, dense rocky matter that surrounds the core with a radius of about 2885 km. It is basically composed of silicate rock rich in iron and magnesium. The mantle is less dense than the core but denser than the outer crust layer.

- The inner region of the Earth is referred to as its core and is divided into a liquid outer core and a solid innermost core of a radius of about 1200 km. The outer core is made up of an iron and nickel alloy, while the inner core is almost entirely made up of solid iron.



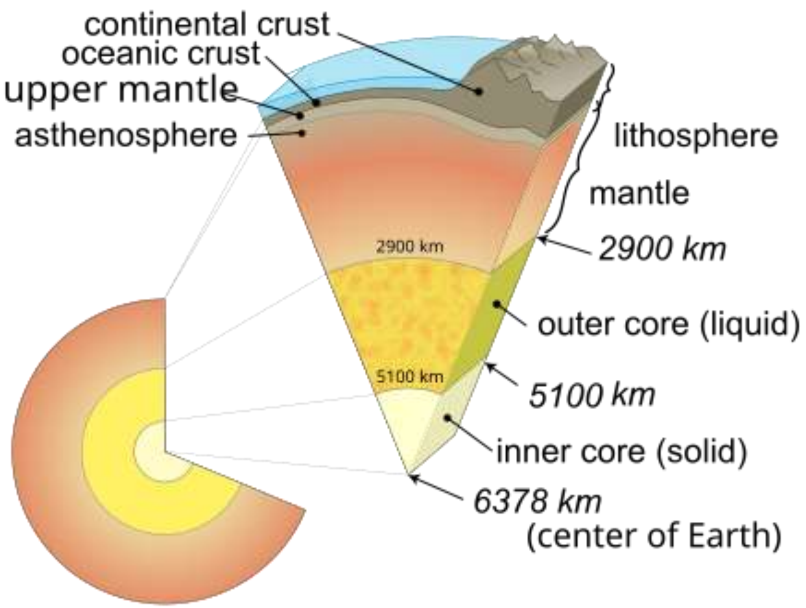
By mass, the Earth is composed mostly of 35% iron, 30% oxygen, 15% Si, 13% Mg, and 7% other elements.

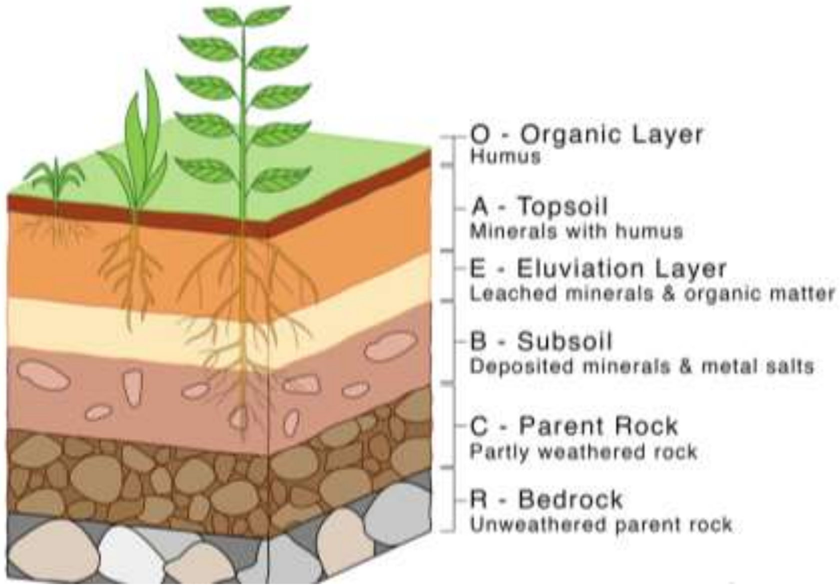
Lithosphere and Soil Profile

- The lithosphere is the rigid, outermost shell of Earth. It is composed of the crust and the upper mantle.
- Its thickness varies between **100 – 200 km** and is made up of rocks and minerals mainly.
- Two types of crust are mainly found in the lithosphere:

➢ **Continental crust** – thick and less dense

➢ **Oceanic crust** – thin and denser





A **soil profile** is a vertical section of soil that shows different layers, or **horizons**.

- O-Horizon (Organic Layer): **decomposed plant and animals (humus)**
- A-Horizon (Topsoil): **Rich in minerals and organic matter, fertile soil**
- B-Horizon (Subsoil): **clay and minerals, less organic matter**
- C-Horizon (Parent Rock): **weathered rock fragments**
- R-Horizon (Bedrock): **Solid rock under soil layers**

Soil Contamination

Soil contamination is the presence of harmful chemicals, toxic substances, or other pollutants in soil at concentrations high enough to affect plants, animals, humans, and the environment.

Sources of Soil Contamination

- **Industrial waste:** Disposal of chemical wastes, leakage of heavy metals like lead (Pb), mercury (Hg), cadmium (Cd), and Effluents from factories. Industries such as tanneries, textile mills, fertilizer plants, thermal power plants, and chemical factories contribute significantly.

- **Modern agricultural practices:** Excessive use of fertilizers (*Urea*, *DAP*, *etc.*), intensive irrigation (*increase salinity and alkalinity*), overuse of pesticides and insecticides, herbicides containing toxic chemicals, etc. The macronutrients accumulate in the soil and enter the food chain.

- **Solid waste:** Dumping of municipal and industrial solid waste, dry garbage, wet garbage, human and animal excerta, biomedical waste, electronic waste, and improper landfill management causes toxic substances to leach into soil. It causes a change in soil color and composition.
- **Precipitation through polluted atmosphere:** The air pollutants like SO_2 and NO_x may fall down on the earth and react with water, and can make the soil acidic. The acidified soil has a very low pH and facilitates the easy mobilization of heavy metals. It also causes the leaching of minerals from the soil and heavily affects trees & plants.
- **Other sources:** Mining activities (release of heavy metals, waste rock deposits, etc.), urbanization, construction, radioactive waste, etc.

Radioactive Pollution

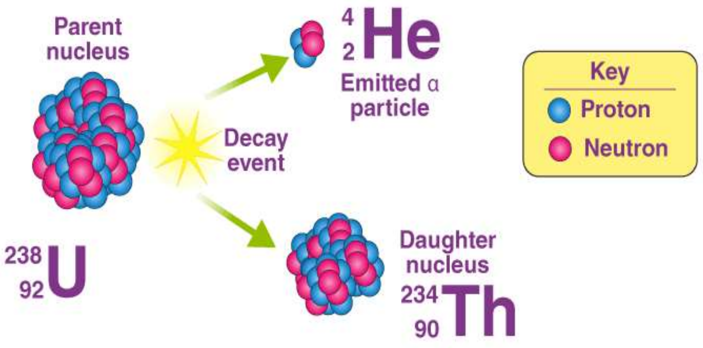
Radioactive pollution refers to the presence of radioactive substances in the environment - air, water, or soil - at levels that are harmful to living organisms. These substances emit **ionizing radiation** (alpha, beta, gamma rays), which can damage biological tissues and disrupt ecological balance.

Sources of Radioactive Pollution:

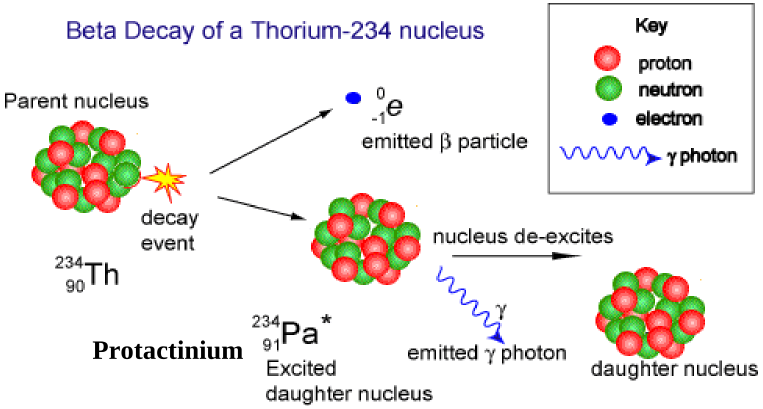
- **Natural Sources:** Cosmic rays from outer space, naturally occurring radioactive materials (uranium, thorium, radon) in rocks and soil, radon gas released from the Earth’s crust.
- **Artificial Sources:** Nuclear power plants (leakage, accidents), nuclear weapons testing and explosions, medical uses (radiotherapy, nuclear medicine waste), industrial uses (radiography, gauges), improper disposal of radioactive waste, mining and processing of nuclear fuels, etc.

Types of Radiation:

Alpha (α) particles – The release of this radiation from the nucleus of a radioactive element is equivalent to the release of a positively charged helium nucleus, which consists of 2 neutrons and 2 protons. Its release results in the loss of 4 atomic mass units from the parent atom. ***Atomic mass = no. of neutrons + protons***, ***Atomic no. = no. of protons in the nucleus***. *Low penetration and can be stopped by a thin sheet or paper, low velocity, and less harmful.*



Beta (β) particles – This is due to the release of electrons from a radioactive nucleus, caused by the break-up of a neutron into a proton and an electron. Thus, it results in an increase in proton, i.e., atomic no. increases. But the mass of released electrons is very small, so the atomic mass will remain the same. *Moderate penetration, -ve charged, high velocity, and can penetrate thin aluminum sheet.*



Gamma (γ) rays – They are electromagnetic radiation that has no charge or mass. It is a very high-frequency photon and can travel at the speed of light. *Short wavelength, high penetration, most dangerous.*

Measuring Units of Radiation

Radioactivity The rate of radiation released by a material	**SI Unit:**	becquerel (Bq)	1 disintegration per second
	Common Unit:	curie (Ci)	3.7×10^{10} disintegrations per second
Exposure The amount of ionizing radiation traveling through the air	**SI Unit:**	$\frac{\text{coulomb}}{\text{kilogram}}$ (C/kg)	3,880 R
	Common Unit:	roentgen (R)	0.000258 C/kg
Absorbed dose The amount of radiation absorbed by an object or person	**SI Unit:**	gray (Gy)	100 rad = 1 J/kg
	Common Unit:	rad	0.01 Gy
Dose equivalence Combination of the amount of radiation absorbed **and** its equivalent health effects	**SI Unit:**	sievert (Sv)	100 rem
	Common Unit:	rem	0.01 Sv

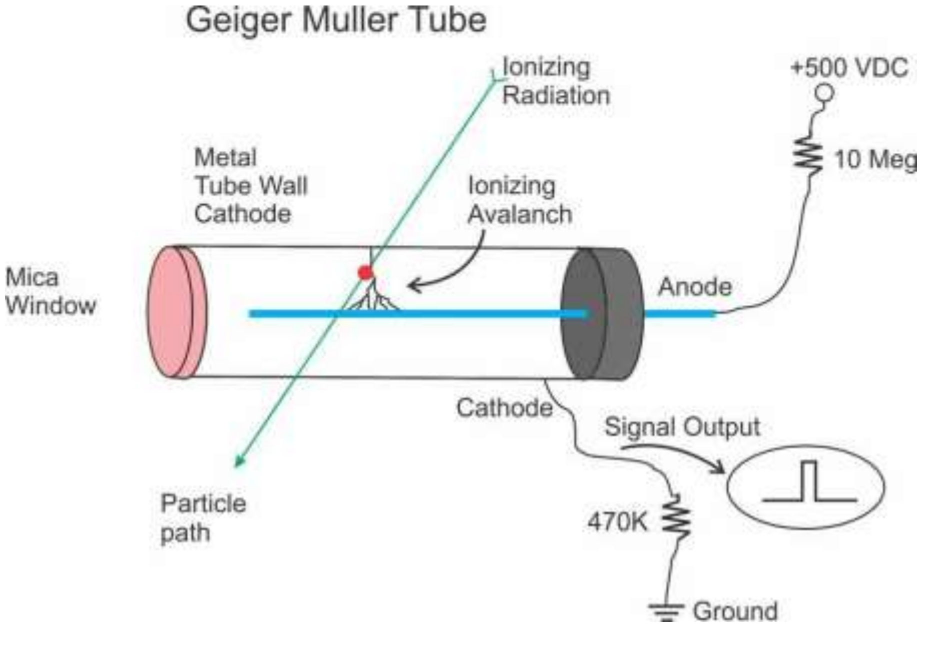
Radiation Measuring Instruments

- **Geiger-Müller (GM) Counter**: This is the most widely used tool for detecting and measuring ionizing radiation. It works by using a gas-filled tube that produces electrical pulses when radiation passes through it. Particle measurements are reported in different units; one of the most widely used is Counts Per Minute (CPM).

GM Counter

Working Principle: The GM Tube is filled with low-pressure noble gas (like Neon or Argon). The outer wall acts as a cathode, and a thin wire runs down the center as the anode. Then, a high voltage, approximately 450-1000 volts, is applied between the wire and the tube wall, creating a strong electric field. The ionization process occurs when radiation (alpha, beta, or gamma) passes through the tube. It strikes gas atoms, knocking out electrons (*ionization*), leaving positively charged ions.

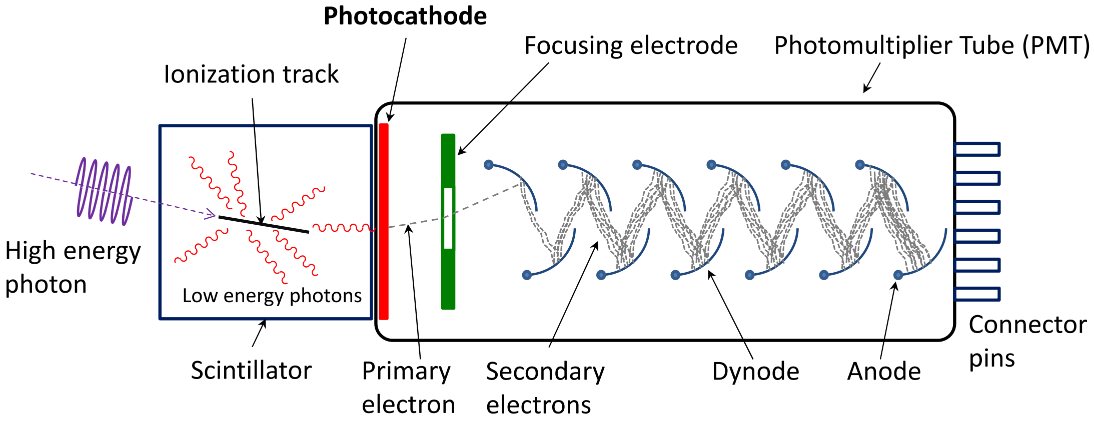
The high voltage accelerates these free electrons toward the center wire. These accelerating electrons hit other gas atoms, causing a cascading "avalanche" of millions of electrons. This massive surge of electrons strikes the wire, producing a momentary electrical current.



- **Dosimeter:** A small, wearable device used to track the total amount of radiation a person has been exposed to over time. Uses a silicon diode detector, a small battery, and a digital display. Radiation strikes the detector, generating electric pulses, and the device amplifies and counts them to calculate the dose rate and accumulated dose.

Dosimeter

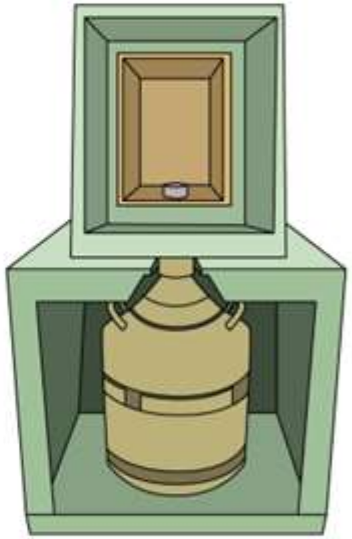
- **Scintillation:** A scintillation detector is a radiation detection device that uses a special material (*scintillator made of inorganic crystals, organic liquids, or plastic*) to convert ionizing radiation into flashes of light. This light is amplified by a photodetector, usually a photomultiplier tube (PMT), into electrical signals for counting or energy measurement.



Scintillation

- **Ionization Chamber**: These are often used for more precise measurements of radiation dose rates in environments like hospitals or nuclear plants

Various Measuring Instruments



Ge Semiconductor Detector

Used to measure radioactivity in foods or soil; Effective in measuring low levels of radioactivity concentrations

NaI (Tl) Food Monitor

Suitable for efficient radioactivity measurement of foods, etc.

Whole-body Counter

Assess accumulation of γ-ray nuclides in the body using numerous scintillation counters or the like

Integrating Personal Dosimeter

Worn on the trunk of the body for 1-3 months to measure cumulative exposure doses during that period

Electronic Personal Dosimeter

Equipped with a device to display dose rates or cumulative doses during a certain period of time and thus convenient for measuring and managing exposure doses of temporary visitors to radiation handling facilities

Effects of Radioactive Pollution

On Human Health

- Radiation sickness
- Cancer (leukemia, thyroid cancer, lung cancer)
- Genetic mutations and birth defects
- Damage to organs and tissues
- Increased mortality at high exposure levels

On Environment

- Soil infertility and contamination
- Water pollution affects aquatic life
- Bioaccumulation and biomagnification in food chains
- Long-term ecosystem imbalance